PONMATHISH T

DAY - 06

1. What is IPv4?

Introduction:
IPv4 stands for **Internet Protocol version 4**. It is a network-layer protocol used to identify devices and deliver data between networks.

Key Points:

- IPv4 uses a **32-bit address**.
- It is divided into **four octets**.
- Each octet ranges from **0 to 255**.
- Example: **192.168.1.10**
- IPv4 supports communication between devices across local networks and the Internet.

Conclusion:
IPv4 is an important protocol that provides addressing and routing for network communication.

2. How many bits are in an IPv4 address?

Introduction:
An IPv4 address uses a fixed number of binary bits to identify a device or network.

Key Points:

- An IPv4 address contains **32 bits**.
- These 32 bits are divided into **4 groups**.
- Each group contains **8 bits**.
- Each 8-bit group is called an **octet**.
- Example: **192.168.1.10**

Conclusion:
Therefore, every IPv4 address consists of **32 bits or 4 octets**.

3. What is an octet?

Introduction:
An octet is a basic unit used to represent an IPv4 address.

Key Points:

- An octet contains **8 bits**.
- Each bit can have a value of 0 or 1.
- An octet can represent values from **0 to 255**.
- IPv4 has four octets.
- Example: In **192.168.1.10**, 192, 168, 1, and 10 are the four octets.

Conclusion:
An octet is simply a group of **8 bits** used in IPv4 addressing.

4. How many octets are in an IPv4 address?

Introduction:
IPv4 addresses are written in dotted-decimal notation.

Key Points:

- An IPv4 address contains **4 octets**.
- Each octet contains **8 bits**.
- Therefore, 4 × 8 = **32 bits**.
- The octets are separated by periods (.).
- Example: **192.168.1.10**

Conclusion:
An IPv4 address contains **4 octets and 32 bits** in total.

5. What is a subnet mask?

Introduction:
A subnet mask is used with an IP address to determine the network and host portions.

Key Points:

- It identifies the **network portion** of an IP address.
- It also identifies the **host portion**.
- A common subnet mask is **255.255.255.0**.
- This is also represented as **/24**.
- Subnet masks help divide networks into smaller subnetworks.

Conclusion:
A subnet mask is essential for understanding network structure and determining whether devices belong to the same network.

6. Difference between network portion and host portion?

Introduction:
An IPv4 address contains a network portion and a host portion.

Network Portion:

- Identifies the network.
- Devices on the same network share the same network portion.
- Used by routers for routing.

Host Portion:

- Identifies a specific device within the network.
- Each host normally needs a unique host address.
- Used to identify individual devices.

Example:
For **192.168.1.10/24**:

- Network portion: **192.168.1**
- Host portion: **10**

Conclusion:
The network portion identifies the network, while the host portion identifies a device within that network.

7. What is a private IPv4 address?

Introduction:
Private IPv4 addresses are used for devices inside private networks such as homes, schools, and organizations.

Key Points:

- They are not directly routable on the public Internet.
- They are commonly used on LANs.
- Routers can use **NAT** to allow private devices to access the Internet.

- They improve efficient use of IPv4 addresses.
- Examples include **192.168.1.10** and **10.0.0.5**.

Conclusion:
Private IPv4 addresses are mainly used for internal network communication.

8. Write all three private IPv4 ranges.

Introduction:
IPv4 has three reserved private address ranges defined for internal network use.

Ranges:

1. **10.0.0.0/8**
   Range: 10.0.0.0 – 10.255.255.255
2. **172.16.0.0/12**
   Range: 172.16.0.0 – 172.31.255.255
3. **192.168.0.0/16**
   Range: 192.168.0.0 – 192.168.255.255

Conclusion:
These three ranges are reserved for private networks and are commonly used in homes, businesses, and organizations.

9. What is a public IPv4 address?

Introduction:
A public IPv4 address is an address used for communication across the public Internet.

Key Points:

- It is generally globally routable.
- It can identify a network or service on the Internet.
- Internet service providers commonly assign public IP addresses.
- Web servers often use public IP addresses.
- Public IP addresses must be managed carefully for security.

Conclusion:
A public IPv4 address enables communication between a network or device and other networks over the Internet.

10. What is the IPv4 loopback address?

Introduction:
A loopback address is used by a computer to communicate with itself.

Key Points:

- The commonly used IPv4 loopback address is **127.0.0.1**.
- The loopback range is **127.0.0.0/8**.
- It does not represent another device on the network.
- It is useful for testing TCP/IP services.
- Developers and network administrators use it to test local applications.

Example:
Typing 127.0.0.1 in a browser can test a web service running on the same computer.

Conclusion:
The loopback address is mainly used for local network testing and troubleshooting.

11. What is APIPA?

Introduction:
APIPA stands for **Automatic Private IP Addressing**.

Key Points:

- It allows a device to automatically configure an IPv4 address.
- It is commonly used when a DHCP server cannot be reached.
- APIPA addresses belong to **169.254.0.0/16**.
- The address is used for local network communication.
- A 169.254.x.x address can indicate a DHCP configuration problem.

Conclusion:
APIPA provides automatic local IPv4 addressing when normal DHCP configuration is unavailable.

12. What does 169.254.x.x commonly indicate?

Introduction:
The 169.254.x.x address range is associated with APIPA.

Key Points:

- It indicates an address in the **169.254.0.0/16** range.
- Windows may automatically assign this type of address.
- It commonly occurs when the device cannot obtain an address from DHCP.
- It may indicate a problem with the DHCP server or network connection.
- It usually allows only limited local network communication.

Conclusion:
A 169.254.x.x address commonly indicates that the device failed to obtain a normal IP address through DHCP.

13. What is a default gateway?

Introduction:
A default gateway is a device that provides a path from one network to another.

Key Points:

- It is commonly a **router**.
- It forwards traffic destined for other networks.
- Home networks commonly use the router as the default gateway.
- A computer sends Internet-bound traffic to its default gateway.
- Without a correct gateway, communication outside the local network may fail.

Example:
A common home gateway might be **192.168.1.1**.

Conclusion:
The default gateway is essential for communication between a local network and external networks.

14. Static vs Dynamic IP addressing?

Introduction:
IP addresses can be assigned manually or automatically.

Static IP

- Configured manually.
- Usually remains unchanged.
- Useful for servers, printers, and network devices.
- Requires manual configuration.

Dynamic IP

- Automatically assigned.
- Usually provided by **DHCP**.
- Can change over time.
- Easier to manage on large networks.

Conclusion:
Static addressing provides consistency, while dynamic addressing provides easier automatic network management.

15. What is DHCP?

Introduction:
DHCP stands for **Dynamic Host Configuration Protocol**. It automatically provides network configuration to devices.

Key Points:

- Assigns an **IP address**.
- Provides a **subnet mask**.
- Provides a **default gateway**.
- Can provide **DNS server information**.
- Reduces the need for manual network configuration.

Conclusion:
DHCP makes network administration easier by automatically configuring devices with the information they need to communicate.

16. What is a network address?

Introduction:
A network address identifies a particular IPv4 network.

Key Points:

- It represents the network rather than an individual host.
- It is determined using the IP address and subnet mask.
- It is used by routers and network administrators.
- A network address cannot normally be assigned to a regular host.
- Example: **192.168.1.0/24** has network address **192.168.1.0**.

Conclusion:
A network address identifies the network to which a group of devices belongs.

17. What is a broadcast address?

Introduction:
A broadcast address is used to communicate with all hosts within an IPv4 network.

Key Points:

- It represents all hosts in a particular subnet.
- It is the highest address in a traditional IPv4 subnet.
- Routers normally do not forward local IPv4 broadcasts.
- Example: For **192.168.1.0/24**, the broadcast address is **192.168.1.255**.
- It can be used when information needs to reach all devices on a local subnet.

Conclusion:
A broadcast address allows communication with all devices within a particular IPv4 subnet.

18. Why is IPv4 knowledge important in cybersecurity?

Introduction:
IPv4 knowledge is essential for cybersecurity because network communication depends heavily on IP addressing.

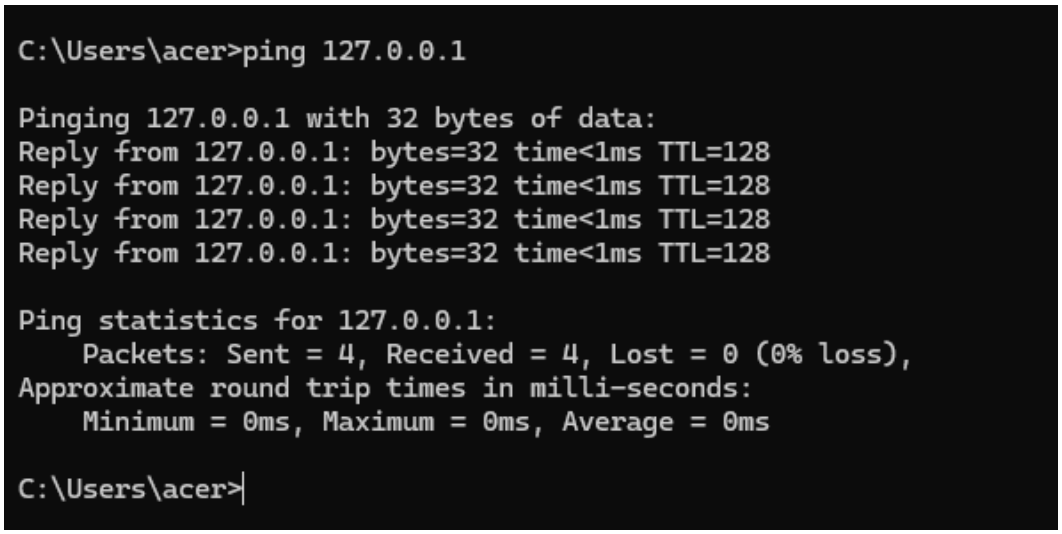
Key Points:

1. **Device Identification:**
   IP addresses help identify devices communicating on a network.
2. **Network Monitoring:**
   Security professionals analyze IP addresses to monitor network traffic.

3. **Threat Detection:**
   Suspicious or unauthorized IP addresses can help identify potential attacks.
4. **Firewall Configuration:**
   IPv4 knowledge helps create rules to allow or block specific addresses and networks.
5. **Incident Investigation:**
   IP addresses can help security teams investigate suspicious network activity and security incidents.

Conclusion:
Therefore, IPv4 knowledge is important for **network monitoring, threat detection, firewall configuration, troubleshooting, and cybersecurity investigations**.

Item	What you need to enter
IPv4 Address	10.150.34.91
Subnet Mask	255.255.255.0
Default Gateway	10.150.34.120
DNS Server	10.150.34.120
DHCP Status	Enabled



```
C:\Users\acer>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : PM
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 5A-E[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

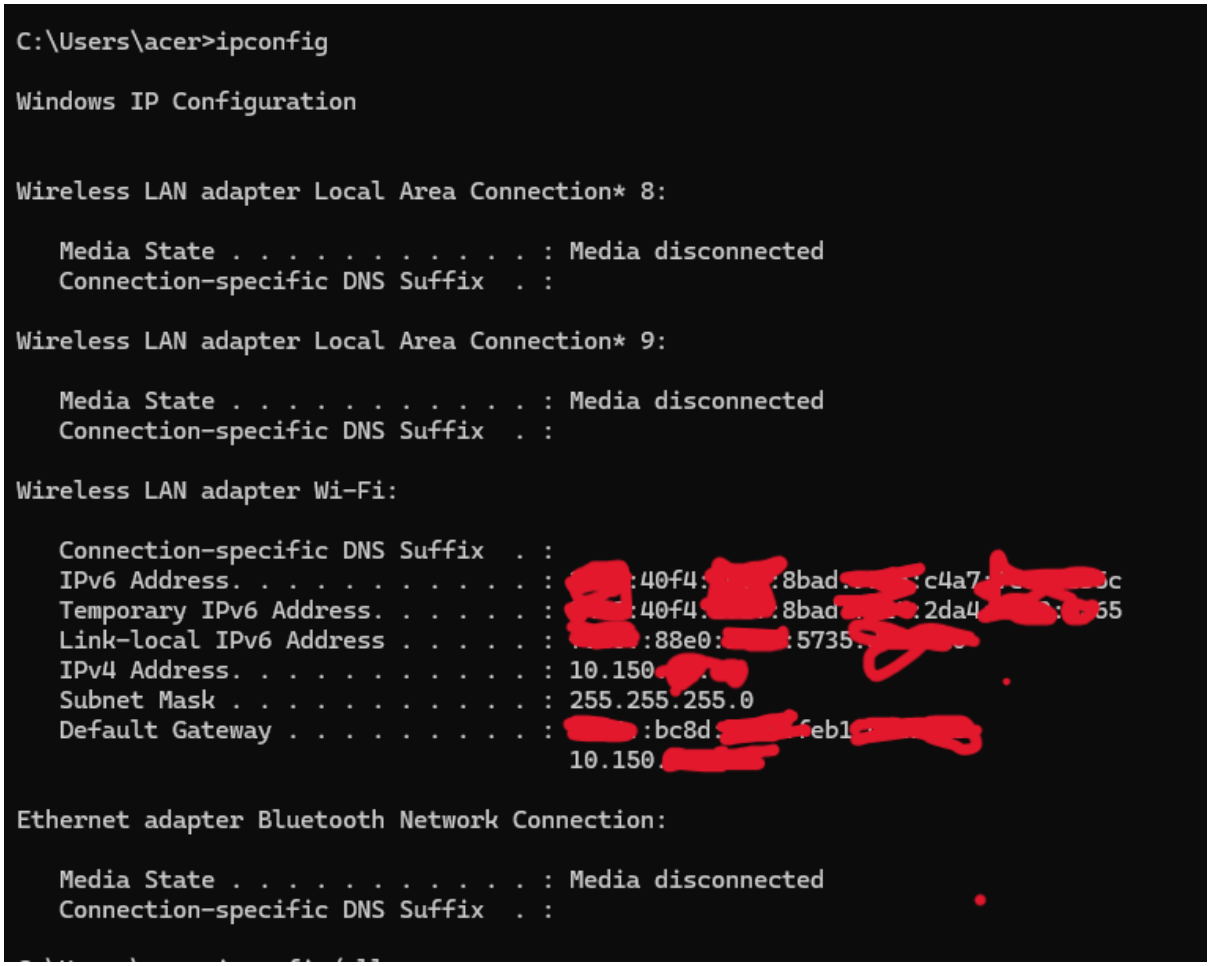
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsof[redacted]irect Virtual Adapter #2
   Physical Address. . . . . . . . . : DA-E[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek 8821CE Wireless LAN 802.11ac PCI-E NIC
   Physical Address. . . . . . . . . : 58-E4-EB-41-04-F2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2409:40f4:[redacted](Preferred)
   Temporary IPv6 Address. . . . . . : 2409:40f4:10[redacted]55(Preferred)
   Link-local IPv6 Address . . . . . : fe80::88e[redacted]referred)
   IPv4 Address. . . . . . . . . . . : 10.150.94.94(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 29 October 2026 07:52:08
   Lease Expires . . . . . . . . . . : 29 October [redacted]
   Default Gateway . . . . . . . . . : fe80::bc8d:[redacted]
                                       10.150.[redacted]
   DHCP Server . . . . . . . . . . . : 10.150.[redacted]
   DHCPv6 IAID . . . . . . . . . . . : 173597931
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-A0-86-E6[redacted]
   DNS Servers . . . . . . . . . . . : 10.15[redacted]
                                       2409[redacted]
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 58-E[redacted]
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```


C:\Users\acer>ipconfig
Windows IP Configuration
Wireless LAN adapter Local Area Connection* 8:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 9:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . :
Temporary IPv6 Address. . . . . :
Link-local IPv6 Address . . . . . :
IPv4 Address. . . . . : 10.150.
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
10.150.
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :